

MEILING WU

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PhD candidate in Operations and Decision Technologies building optimization, machine learning, and forecasting tools for decisions under uncertainty, with applications in higher education, supply chain, and healthcare.

RESEARCH EXPERIENCE

University Admissions Enrollment Decision-Support System, PhD Research, UC Irvine 2023–Present

- **Product:** Identified university relied on manual judgment amid over- and under-enrollment in past cycles; designed a **decision-support system** to meet enrollment targets while balancing cohort quality, budget, and enrollment risk.
- **ML & Data:** Built applicant-level deposit and enrollment **prediction models** using **XGBoost** and **logistic regression** on 6 years of admissions data, increasing true positives by **27%** and reducing false positives by **23%** versus baseline.
- **Decision-Making:** Integrated applicant-level **forecasts with stochastic dynamic optimization** to select **admit / hold / reject decisions** and update recommendations as new applicant arrivals and deposit information became available.
- **Impact & Stakeholders:** Reduced target-mismatch **cost by \$536K** versus benchmark in out-of-sample **simulation** while meeting seat target and improving cohort quality; presented the system and recommendations to admissions **leadership**.

Agricultural Supply Chain Optimization, PhD Research, Washington University in St. Louis 2021–2022

- **Problem Solving:** Translated planting, crop-maturity, and storage constraints across **1,375+ corn populations** into a capacity-planning problem to stabilize weekly harvest volume under uncertain weather.
- **Forecasting & ML:** Developed and benchmarked **LSTM** and **SARIMA** models on 10 years of weather data to forecast daily temperatures and estimate harvest timing from cumulative heat units, achieving **RMSE = 0.28** and **95%** correlation.
- **Technical Execution:** Integrated weather forecasts and crop attributes into a **Gurobi MILP decision engine** to optimize planting schedules while respecting a 7,000-unit **weekly storage-capacity constraint**.
- **Business Impact:** Reduced weekly harvest-capacity mismatch by **97%** (**223K** → **6.8K** units), eliminating **17 over-capacity** and **35 under-capacity weeks** and projecting **\$1.4M** in **annual operational cost avoidance**.

CONSULTING EXPERIENCE

Fit & Food Connection, Nonprofit, Team Lead, Olin Business Consulting Project, WUSTL Fall 2020

- **Leadership & User Insights:** Led a **4-person consulting team** to analyze volunteer feedback and identify onboarding friction and uneven workloads as key drivers of volunteer attrition.
- **Process Design:** Redesigned the onboarding experience with standardized SOPs, digital training, assessments, and a buddy program, cutting onboarding time from **2 weeks to 3 days**.
- **Impact:** Introduced workload tracking, career-development incentives, and team-building initiatives, increasing **average volunteer attendance by 41%**.

HybridHelix Consulting LLC, Consultant Intern Summer 2020

- **Growth:** Improved organic customer acquisition through SEO and website optimization, achieving **Google first-page rankings for 16 target search terms**.
- **ML & Customer Insights:** Built and evaluated **logistic regression and random forest** models to predict lead-response likelihood, achieving approximately **90% validation accuracy** and improving outreach prioritization.
- **Data Infrastructure:** Consolidated CRM data across leads, touchpoints, and outreach history to enable **customer segmentation, funnel analysis, and conversion tracking**.

Edward Jones, Olin Business Consulting Project, WUSTL Spring 2020

- **Opportunity Assessment:** Evaluated accounting workflows across time savings, FTE impact, error reduction, implementation cost, and operational risk to identify **high-value automation opportunities**.
- **Business Modeling:** Built a **multi-scenario, multi-year ROI model** comparing implementation costs with labor-capacity savings, identifying **\$1M+ in annual capacity value** and an **18–24 month payback period**.
- **Prioritization & Stakeholders:** Presented a phased automation roadmap to client leadership, prioritizing **high-ROI, low-risk processes** for initial implementation.

PRESENTATION AND AWARD

Presented the paper “*Admit or Wait: Dynamic Admission Control under Non-Stationary Arrivals*” at INFORMS Annual Meetings (2024, 2025) and the DSI Annual Conference (2025). **Reward: Ray Watson Doctoral Fellowship**

EDUCATION

University of California, Irvine, Paul Merage Business School 2021-2026

- **PhD in Operations and Decision Technologies**, GPA: 3.9/4.0, Ray Watson Doctoral Fellowship, Full Scholarship
- Relevant Coursework: Optimization | Dynamic Programming | Inventory Control | Applied Econometrics | Causal Inference

Washington University in St. Louis, Olin Business School 2019-2021

- **Master of Science in Supply Chain Management**, GPA: 3.8/4.0
- Relevant Coursework: Project Management | Operations Management | Machine Learning in R | Deep Learning in Python

Tianjin Normal University, Business School 2013-2017

- **Bachelor of Science in Logistics Management**, GPA: 3.3/4.0

LEADERSHIP & SKILLS

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- **Vice President**, Olin Supply Chain & Operations Association, Washington University in St. Louis 2020- 2021

Built **partnerships and organized panel discussion and lab visit with local firms** (Bayer, Budweiser) for WUSTL students.
Advertised through social media which increased **attendance over 100s** and **raised funds by 400%** compared to last year.

- **Co-Founder & President**, Bilingual Public Speaking Club, Tianjin Normal University (China) 2014-2017

Grew to **70+** members (Chinese and international students) ; led weekly prepared speeches and impromptu speaking.
Coached 3 students prepare for “The China Daily National English Speaking Competition”; 1 won Third Prize at state level.